

OLIST MARKETPLACE PERFORMANCE DASHBOARD

Solution Report

Client	Ana Ferreira, Head of Marketplace Operations, Olist
Prepared for	BI Analyst / Dashboard Developer
Project brief date	September 2026
Solution platform	Microsoft Power BI
Dataset	Brazilian E-Commerce Public Dataset by Olist
Solution status	Six-page dashboard implemented; documentation prepared

This report translates the original project brief into the implemented Power BI solution, including the business problem, architecture, data preparation, DAX, dashboard design, findings, limitations and final deliverables.

1. Original Requirement

The project brief requested a single source of truth for Olist operations, sales and logistics teams, replacing slow and error-prone manual spreadsheet reporting. The requested solution needed to explain revenue drivers, delivery performance, customer satisfaction, payment behavior, seller performance and geography.

The brief specifically requires six core dashboard areas plus Seller Performance as a stretch goal, filters for date/state/category, drill-through and a professional design suitable for leadership.

Source: project brief — background, problem statement and objectives. ■filecite■turn5file0■L9-L31■

Required pages and implementation status

Requirement	Implemented solution	Status
Executive Summary	KPIs, monthly revenue trend, order status, sales by year, category performance	Completed
Sales & Product Performance	Revenue/category analysis, price vs freight, top/bottom performance, trend and matrix	Completed
Delivery & Logistics	Actual vs estimated delivery, late/on-time, state delivery, delay vs review score, cancellations	Completed
Customer Satisfaction	Review distribution, state low-rating %, delay vs review score, category review score	Completed
Payments	Payment data loaded/modelled; no dedicated final report page shown	Partial / not a separate page
Geography	Sales/orders by state, delivery time by state, map	Completed
Seller Performance	Seller revenue, orders and revenue/review analysis	Completed (stretch goal)
Slicers	State, Product Category and Year/date context	Completed
Reset control	Reset/Clear Slicer button	Completed
Drill-through	Requested in brief; not verified in final screenshots	Not verified

The original brief's required pages and must-have features are documented at lines 36–50. ■filecite■turn5file0■L36-L50■

2. Proposed Solution

The solution is a Power BI analytical application that combines multiple Olist CSV datasets into a relationship-based model and presents business KPIs through six report pages.

- Create a sales fact structure from Orders + Order Items for revenue, quantity, freight, product and seller analysis.
- Keep Payments and Reviews separate because their grains differ from order-item sales data.
- Create a dedicated Date table for monthly, quarterly, yearly and time-intelligence analysis.
- Use Products, Sellers, Customers and Orders as analytical dimensions/bridges.
- Develop DAX measures for revenue, orders, AOV, delivery performance, reviews and payments.
- Use interactive slicers and a reset control to support self-service analysis.
- Present the final output as a leadership-oriented multi-page Power BI report.

3. Data Sources and Preparation

The project uses the Brazilian E-Commerce Public Dataset by Olist. The original brief identifies Kaggle as the source and lists the nine CSV files used for the project.

Source: project brief — data source and included files. ■filecite■turn5file0■L69-L76■

Dataset	Role in solution
Orders	Order status and lifecycle dates; order-level bridge

Dataset	Role in solution
Order Items	Sales fact grain; product, seller, price and freight
Products	Product and category analysis
Customers	Customer location/state analysis
Sellers	Seller performance analysis
Payments	Payment methods, values and installments
Reviews	Review scores and customer satisfaction
Geolocation	Geographic reference information
Category Translation	Portuguese-to-English category mapping

Cleaning and transformation decisions

- Complete duplicate rows were considered separately from repeated business keys.
- Repeated order_id values in Order Items and Payments were retained because they are valid at those tables' respective grains.
- Product records with valid missing descriptive attributes were retained rather than assigning invented values.
- Review comments were allowed to remain blank where no comment was provided.
- Portuguese category names were translated using the category translation file.
- Date columns were converted to suitable date/date-time types.
- Orders and Order Items were combined into the sales fact structure used by the dashboard.
- Payments and Reviews remained separate to avoid row multiplication.

4. Data Model / Architecture

The model follows a dimensional relationship approach. The sales fact is at order-item grain, while Payments and Reviews remain separate facts connected through the order-level table.

Table	Role	Key analytical use
Date	Dimension	Time intelligence and date filtering
products_dataset	Dimension	Product/category filtering
sellers_dataset	Dimension	Seller filtering and performance
customers_dataset	Dimension	Customer state/location
olist_orders_dataset	Order bridge	Connect sales, payments and reviews
facts_sales	Main fact	Revenue, quantity, freight, delivery
facts_payments_dataset	Fact	Payment value/type/installments
facts_reviews_dataset	Fact	Review score and review analysis

Key relationship design

- Date[Date] → facts_sales[purchase_date]
- Products[product_id] → facts_sales[product_id]
- Sellers[seller_id] → facts_sales[seller_id]
- Customers[customer_id] → olist_orders_dataset[customer_id]
- olist_orders_dataset[order_id] → facts_sales[order_id]
- olist_orders_dataset[order_id] → facts_payments_dataset[order_id]
- olist_orders_dataset[order_id] → facts_reviews_dataset[order_id]

5. DAX Solution

The DAX layer provides reusable measures rather than hard-coded visual calculations. This allows slicers and report filters to dynamically recalculate KPIs.

Measure	Solution logic
Total Sales	SUM(facts_sales[price])
Total Orders	DISTINCTCOUNT(facts_sales[order_id])
Total Quantity	COUNTROWS(facts_sales)
Average Order Value	DIVIDE([Total Sales], [Total Orders])
Sales YTD / MTD / QTD	Time-intelligence functions using Date[Date]
Sales Previous Year	CALCULATE([Total Sales], SAMEPERIODLASTYEAR('Date'[Date]))
On-Time Delivery %	On-time delivered orders / delivered orders
Late Orders	Delivered orders where delivered date > estimated date
Average Delivery Days	Order-level average of purchase-to-delivery duration
Average Review Score	AVERAGE(facts_reviews_dataset[review_score])
Total Reviews	DISTINCTCOUNT(facts_reviews_dataset[review_id])
Total Payment Value	SUM(facts_payments_dataset[payment_value])
Average Payment per Order	Payment value / distinct paid orders

AOV was deliberately calculated as total sales divided by distinct orders because the sales fact is at order-item grain. Using AVERAGE(price) would not represent average order value correctly.

6. Implemented Dashboard Pages

Page	Implemented solution
1. Executive Summary	Total Sales, Total Orders, AOV, Average Review Score, On-Time Delivery %, monthly revenue trend, order status, sales by year and category performance.
2. Sales & Product Performance	Revenue trend, price vs freight cost, top/bottom revenue analysis and Sales Performance Matrix.
3. Delivery & Logistics	Actual vs estimated delivery, delay vs review score, state delivery time, late/on-time delivery and cancelled orders.
4. Customer Satisfaction	Review distribution, low-rating review % by state, delivery delay vs review score and category review scores.
5. Geography Analysis	Sales by state, orders by state, revenue map and delivery days by state.
6. Seller Performance	Seller sales, seller orders and seller revenue/review-score analysis.

The final screenshots show six report pages. The original brief also requested a dedicated Payments page; payment data exists in the model, but a separate Payments page is not shown in the final six-page solution.

7. Solution Results and KPIs

KPI	Observed dashboard value
Total Sales	13.59M
Total Orders	99K
Average Order Value	137.75
Average Review Score	4.09
On-Time Delivery	94.16%
Delivered Orders	96K

KPI	Observed dashboard value
Late Orders	8K
Average Delivery Days	12.41
Total Reviews	98K

These are the values visible in the final dashboard screenshots supplied for the solution report.

8. Business Insights and Recommendations

The original brief requires 3–5 actionable insights with recommendations. ■filecite■turn5file0■L55-L64■

Insight 1 — Revenue is concentrated in leading categories

Finding: The Executive Summary shows leading displayed categories including Health & Beauty (~1.26M), Watches & Gifts (~1.21M), Bed, Bath & Table (~1.04M), Sports & Leisure (~0.99M) and Computers & Accessories (~0.91M).

Recommended action: Recommendation: monitor these categories through recurring category-level sales, order and AOV reporting and investigate changes in their contribution over time.

Insight 2 — Delivery performance varies substantially by state

Finding: Overall on-time delivery is 94.16%, while the state-level chart shows average delivery time ranging from approximately 9 days in SP to approximately 29 days in RR in the displayed report.

Recommended action: Recommendation: use state-level delivery monitoring to investigate long delivery times, carrier constraints and the accuracy of estimated delivery dates.

Insight 3 — Delivery and customer satisfaction should be investigated together

Finding: The delivery-delay vs average-review-score scatter plot shows variation across states in both measures.

Recommended action: Recommendation: use state, category and delivery-delay segments to investigate lower review performance. Treat the relationship as an analytical association rather than proof of causality.

9. Validation and Quality Checks

- Reset/Clear Slicer control was added and tested as part of the report interaction workflow.
- State, Product Category and Year/date slicers were checked across the dashboard pages.
- AOV was checked against the distinct-order calculation.
- Delivery KPIs were reviewed against the completed dashboard visuals.
- Customer state filtering was corrected so state-level review metrics vary rather than remaining constant.
- Category-to-review analysis uses a controlled order-level filtering approach rather than indiscriminately changing relationship directions.
- The Sales Performance Matrix's very large 2017 YoY value was identified as a partial-period comparison issue.

10. Requirement Gap / Final Status

Original requirement	Final solution status	Comment
Single source of truth	Completed	Six-page interactive Power BI report
Leadership-ready design	Completed	Consistent dashboard layout, KPI cards, slicers and navigation
Date/state/category filters	Completed	Interactive slicers included
Reset filters	Completed	Reset/Clear Slicer control included
Delivery delay vs review score	Completed	State-level scatter visual included

Original requirement	Final solution status	Comment
3+ actionable insights	Completed	Three evidence-based insights documented
Payments page	Not in final six-page report	Payment data/model exists, but no dedicated final page is shown
Drill-through	Not verified	Requested in brief; not evidenced in supplied final screenshots

11. Final Deliverables

- Power BI report (.pbix) containing the data model, DAX and six dashboard pages.
- Published Power BI report or screenshots, depending on Power BI Service availability.
- Solution report documenting requirements, implementation, results, insights and limitations.
- README/project documentation for portfolio or GitHub use.
- One-page stakeholder summary based on the final dashboard.

12. Conclusion

The completed Olist Power BI solution transforms multiple e-commerce datasets into a reusable analytical report. It provides a consolidated view of sales, products, delivery, customer satisfaction, geography and sellers, while using DAX measures and a structured relationship model to support interactive analysis.

The final dashboard surfaces three major areas for management attention: revenue concentration in leading categories, geographic variation in delivery performance, and the need to evaluate delivery performance alongside customer satisfaction. The solution therefore addresses the central analytical need described in the original project brief while clearly documenting the remaining scope items that were not evidenced in the final report.

Original project brief reference: Olist Marketplace Performance Dashboard, September 2026. ■filecite■turn5file0■L3-L5■